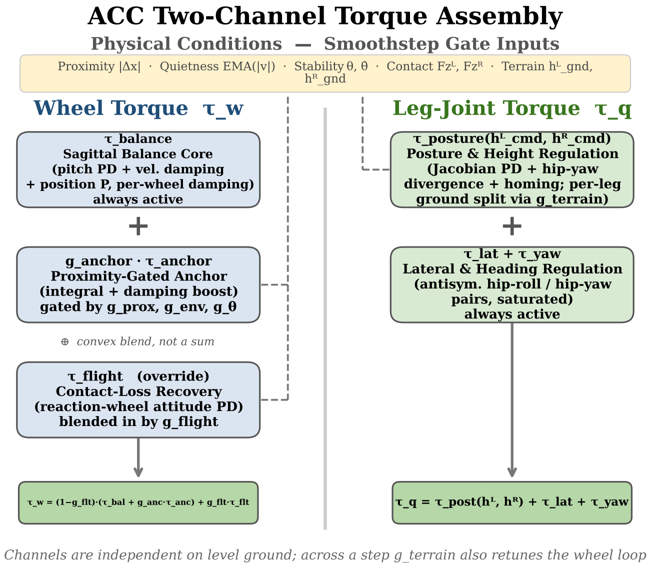
ACC Two-Channel Torque Assembly
Physical Conditions — Smoothstep Gate Inputs
Proximity |Δx| · Quietness EMA(|v|) · Stability θ, θ · Contact Fzᴸ, Fzᴿ · Terrain hᴸ_gnd, hᴿ_gnd
Wheel Torque τ_w
τ_balance Sagittal Balance Core (pitch PD + vel. damping + position P, per-wheel damping) always active
+
g_anchor · τ_anchor Proximity-Gated Anchor (integral + damping boost) gated by g_prox, g_env, g_θ
⊕ convex blend, not a sum
τ_flight (override) Contact-Loss Recovery (reaction-wheel attitude PD) blended in by g_flight
τ_w = (1−g_flt)·(τ_bal + g_anc·τ_anc) + g_flt·τ_flt
Leg-Joint Torque τ_q
τ_posture(hᴸ_cmd, hᴿ_cmd) Posture & Height Regulation (Jacobian PD + hip-yaw divergence + homing; per-leg ground split via g_terrain)
+
τ_lat + τ_yaw Lateral & Heading Regulation (antisym. hip-roll / hip-yaw pairs, saturated) always active
τ_q = τ_post(hᴸ, hᴿ) + τ_lat + τ_yaw
Channels are independent on level ground; across a step g_terrain also retunes the wheel loop